

Automating Software Installation with Shell Scripts

As a DevOps Engineer, automating processes is key to enhancing efficiency. I recently created shell scripts to streamline the installation of Jenkins and Apache Tomcat on a server. This eliminates the need to manually execute multiple commands and ensures a consistent setup every time.

1. Automating Jenkins Installation

The following script updates the system, installs Java, and sets up Jenkins:

```
#!/bin/bash

# Update the system packages
echo "Updating system packages..."
sudo apt update -y && sudo apt upgrade -y

# Install Java (required for Jenkins)
echo "Installing Java..."
sudo apt install openjdk-11-jdk -y

# Verify Java installation
echo "Verifying Java installation..."
java -version

# Install Jenkins
echo "Installing Jenkins..."
wget -q -O - https://pkg.jenkins.io/debian/jenkins.io.key | sudo apt-key add -
sudo sh -c 'echo deb http://pkg.jenkins.io/debian-stable binary/ > /etc/apt/sources.list.d/jenkins.list'
sudo apt update -y && sudo apt install jenkins -y

# Start and enable Jenkins
echo "Starting Jenkins service..."
sudo systemctl start jenkins
sudo systemctl enable jenkins

# Verify Jenkins status
echo "Checking Jenkins status..."
sudo systemctl status jenkins

echo "Jenkins is accessible at http://<your_server_ip>:8080"
echo "Jenkins installation is complete."
```

2. Automating Tomcat Installation

To avoid port conflicts with Jenkins, the Tomcat server is set to run on port 8091. Here's the script:

```
#!/bin/bash

# Update the system packages
echo "Updating system packages..."
sudo apt update -y && sudo apt upgrade -y

# Install Java (required for Tomcat)
echo "Installing Java..."
sudo apt install openjdk-11-jdk -y

# Verify Java installation
echo "Verifying Java installation..."
java -version

# Install Apache Tomcat
echo "Installing Tomcat..."
TOMCAT_VERSION=9.0.73
sudo mkdir -p /opt/tomcat
Wget
https://archive.apache.org/dist/tomcat/tomcat-9/v$TOMCAT_VERSION/bin/apache-tomcat-$TOMCAT_
VERSION.tar.gz -P /tmp
sudo tar -xvzf /tmp/apache-tomcat-$TOMCAT_VERSION.tar.gz -C /opt/tomcat
sudo ln -s /opt/tomcat/apache-tomcat-$TOMCAT_VERSION /opt/tomcat/latest
sudo chown -R $(whoami):$(whoami) /opt/tomcat
sudo chmod +x /opt/tomcat/latest/bin/*.sh

# Change Tomcat port to 8091
echo "Updating Tomcat to run on port 8091..."
sudo sed -i 's/port="8080"/port="8091"/' /opt/tomcat/latest/conf/server.xml

# Start Tomcat
echo "Starting Tomcat server..."
/opt/tomcat/latest/bin/startup.sh

echo "Tomcat installed and running on port 8091"
echo "You can access Tomcat at http://<your_server_ip>:8091"
echo "Tomcat installation is complete."
```

Instructions:

1. Save each script as `install_jenkins.sh` and `install_tomcat_8091.sh`.
2. Make the scripts executable:

```
chmod +x install_jenkins.sh
chmod +x install_tomcat_8091.sh
```

3. Run the scripts as `sudo`:

```
sudo ./install_jenkins.sh
sudo ./install_tomcat_8091.sh
```

4. Replace `<your_server_ip>` with the actual IP address of your server.

Summary

- **Jenkins** will be installed and accessible at `http://<your_server_ip>:8080`.
- **Tomcat** will be installed and accessible at `http://<your_server_ip>:8091`, with the default port changed to avoid conflict with Jenkins.

Changing Tomcat from 8080 to 8091 (or another port) can be necessary for several reasons:

1. **Port Conflicts:** If another service is using port 8080, changing to 8091 prevents conflicts.
2. **Security:** Non-standard ports can reduce exposure to automated attacks targeting default ports.
3. **Environment Requirements:** Some deployments might require specific ports due to policies or architecture.
4. **Multiple Instances:** Running multiple Tomcat instances on the same server requires different ports.
5. **Load Balancer Configuration:** Adjusting the port can align with a load balancer or reverse proxy setup.

Summary

You should change the port only if needed, but you can also configure Tomcat to run on both ports (8080 and 8091) for flexibility.